Session 11: Capstone Project Planning and Mission Definition

Task 1: Mission Statement & Measurable Objectives

1. Mission Statement:

Provide continuous, high-revisit Earth observation coverage over the entire landmass and maritime zones of Vietnam for a 5-year operational lifespan to support disaster monitoring, maritime domain awareness, and environmental management.

2. Measurable Mission Objectives

- **Objective 1 (Revisit Time & Coverage):** Achieve a mean revisit time of **less than 30 minutes** for any location within the geographic rectangle defined by latitudes $8.5^o N - 23.5^o N$ and longitudes $102.0^o E - 118.0^o E$.
- **Objective 2 (System Scale & Orbit):** Deploy and maintain the full operational constellation within a budget of **no more than 24 small satellites operating** in Sun-Synchronous Orbit (SSO) at an altitude of $500\ km \leq h \leq 600\ km$.
- **Objective 3 (Lifetime & Space Debris Mitigation):** Ensure an operational lifetime of at least **5 years**, with a mandatory controlled post-mission disposal capability to re-enter the atmosphere within **less than 2 years** after end of life.

Task 2: Requirements & Verification Flowdown

1. Verifiable Requirements Table

Req ID	Requirement Statement ("Shall")	Parent Objective	Verification Method
SYS-REQ-01	The satellite constellation **shall** provide a mean revisit time below 30 minutes over the latitude bounds $8.5^o N - 23.5^o N$	Objective 1	**Simulation** (STK / Coverage Analysis)
SYS-REQ-02	Each satellite **shall** maintain its operational orbit within a $\pm 5\ km$ altitude deadband for a period of 5 years.	Objective 1 + 2	**Analysis & Simulation**
SYS-REQ-03	The onboard propulsion subsystem **shall** provide a minimum total Δv capacity of 300 m/s per satellite.	Objective 2 + 3	Analysis (Mass & Propulsion Budget)

SYS-REQ-04	Each satellite **shall** execute a de-orbit maneuver at EOL to reduce perigee altitude ensuring atmospheric re-entry within 2 years.	Objective 3	**Analysis** (Orbit Propagation)
SYS-REQ-05	The optical payload **shall** support an off-nadir roll steering angle of at least $\pm 30^{o}$ to enhance cross-track target access.	Objective 1	**Inspection & Simulation**
SYS-REQ-06	The maximum wet mass of each satellite at launch injection **shall** not exceed 150 kg.	Objective 2	**Analysis / Test** (Mass Measurement)

2. Requirements Flowdown Diagram

Mission Objective 1: Revisit < 30 min

↓ (Top-Down Traceability)

System Req: SYS-REQ-01 (Coverage) & SYS-REQ-05 (Roll Steering $\pm 30^{o}$)

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Subsystem Req: ADCS Roll Control Torque & Optical FOV Capabilities

Task 3: Concept of Operations (ConOps)

Phase	Description	Key Astrodynamics Needs & Maneuvers
Phase 1: Launch & Ascent	Co-ride / Dedicated launch vehicle insertion into initial deployment orbit.	Launch vehicle trajectory evaluation, separation vector modeling, injection error analysis.
Phase 2: Deployment (LEOP)	Satellite separation, solar array deployment, orbit phasing into Walker planes.	Initial orbital phasing maneuvers ($\Delta v_{phasing}$), differential drag or impulsive thruster burns
Phase 3: Commissioning	Payload calibration, ADCS detumbling, cross-link and ground communication checks.	Orbit determination (OD) refinement, initial attitude alignment to nadir-pointing.
Phase 4: Operations	5-year primary mission lifecycle for imaging coverage over Vietnam.	Drag make-up station-keeping burns (Δv_{SK}), inclination drift

		management, collision avoidance maneuvers.
Phase 5: Disposal / EOL	De-commissioning and post-mission disposal.	De-orbit retro-burn ($v_{deorbit}$) to lower perigee into dense atmosphere for rapid re-entry.

Task 4: Trade Space, Technical Budgets & Margins

1. Trade Space & Figures of Merit (FOM)

- **Design Variables** (x)**:** Altitude $(500 - 600\ km)$, Inclination $(i = 97.4^o - 97.8^o)$, Total satellites $(N = 12 - 24)$, Planes $(P = 3 - 6)$, Off-nadir angle $(\pm 20^o - \pm 30^o)$.
- **Figures of Merit (FOM):** Mean revisit time over Vietnam, maximum coverage gap, total delta-v requirement, satellite mass, constellation cost

2. Lifetime Delta-v Budget (Per Satellite, 500 km Altitude, 5-year Life)

Maneuver Item	Required Δv (Nominal)
Orbit Injection Error Correction & Phasing	$40.0\ m/s$
Station-Keeping / Drag Make-Up (5 years)	$50.0\ m/s$
Collision Avoidance Maneuvers (CAM)	$10.0\ m/s$
Post-Mission Disposal (Perigee lowering to $< 250\ km$)	$110.0\ m/s$
Subtotal (Nominal)	$210.0\ m/s$
Contingency (+20%)	$42.0\ m/s$
Total Required Budget	$252.0\ m/s$

3. Margin Calculation

- **Propulsion Capability Delivered**: $J_C = 300.0\ m/s$
- **Requirement Budget**: $J_R = 252.0\ m/s$

$$Margin = \frac{Capability - Requirement}{Requirement} \times 100\% = \frac{300.0 - 252.0}{252.0} \times 100\%$$
$$= 19.05\%$$

Task 5: Risk Assessment & Risk Matrix

1. Identified Risks

- **R1 (Launch Vehicle Shortfall):** Launch delay or insertion under-performance leaves planes incompletely populated $(L = 4, C = 4 \Rightarrow R = 16, High/Red)$

- **R2 (Enhanced Atmospheric Drag):** Solar maximum conditions increase upper atmosphere density, depleting station-keeping fuel early $(L = 3, C = 3 \Rightarrow R = 9, Medium/Amber)$
- **R3 (Single Satellite In-Orbit Failure):** Early operational loss of one spacecraft causing localized coverage gaps $(L = 2, C = 3 \Rightarrow R = 6, Medium/Amber)$

2. 5x5 Risk Matrix

Lightlihood (L) \ Consequence (C)	**1 (Insignificant)**	**2 (Minor)**	**3 (Moderate)**	**4 (Major)**	**5 (Catastrophic)**
5 (Almost Certain)	Low	Medium	High	High	High
4 (Likely)	Low	Medium	Medium	**R1 (High)**	High
3 (Possible)	Low	Low	**R2 (Medium)**	Medium	High
2 (Unlikely)	Low	Low	**R3 (Medium)**	Medium	Medium
1 (Rare)	Low	Low	Low	Low	Medium

3. Mitigation Strategies

- **Mitigation for R1 (Red Risk**): Implement a phased deployment strategy where initial operational capability (IOC) is achieved with 12 satellites, supplemented by ground-stored backup units.
- **Mitigation for R2 (Amber Risk):** Carry a minimum 19.05% delta-v margin in propulsion sizing to absorb atmospheric density variations.
- **Mitigation for R3 (Amber Risk):** Design orbital planes with flexible phasing parameters to redistribute remaining active satellites across the plane if an in-orbit failure occurs.